Matthew **Maisano**

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About Me

As a Physics graduate and current Master's student in Theoretical Physics, I have always been driven by a profound curiosity about the scientific world. For the past five years, I have shared this passion by tutoring university and high school students in math and physics. I consider myself an enthusiast with diverse interests, always eager to explore new scientific frontiers and the complex mechanisms that govern our universe.

Education

University of Milano - Bicocca *Milan, IT*

MASTER OF THEORETICAL PHYSICS *October 2023 - September 2026*

- Master Thesis (Not Finished Yet) : Black Holes with homogeneous horizon in gauged supergravity
- Courses: Quantum Field Theory I & II, Mathematical Methods for Physics, General Relativity, Statistical Mechanics, Theoretical Physics I & II, Quantum Gravity, Laboratory of Computational Physics, Topological Methods in Field Theories
- Extra Courses Followed: Optimal Control Theory, Geometrical Methods for Physics, Phenomenology Theory

University of Milano - Bicocca *Milan, IT*

BACHELOR OF PHYSICS *October 2020 - November 2023*

- Bachelor Thesis : Instantons in Quantum Mechanics and Quantum Field Theory
- Courses: Mathematical Analisis I, Linear Algebra and Geometry, Chemistry, Laboratory I, Physics I, Laboratory of Computer Science, Classical Mechanics, Mathematical Analysis II, Laboratory II, Physics II, Physics III, Mathematics for Physics, Elements of Astrophysics, Quantum Mechanics, Structure of Matter,Plasma Physics Laboratory,Geometry III, Nuclear and Subnuclear Physics, Relativity, Computing Skills.
- Extra Courses followed: Geometry I, Geometry II.

Skills

Technical Skills

- Python
- C & C++
- Mathematica
- ROOT

Soft Skills

- Descriptive and mathematical modeling
- Analytical thinking and complex problem solving
- Proactivity and leadership
- Autonomy and self-management

Experience

WORK ACTIVITIES: *Busto Arsizio, Italy*

June 2020 - Currently Working

- Event Assistant & Coordinator | Erba Fina (3-year tenure): Managed the logistical and technical execution of private and corporate events. I gained significant experience in multitasking and problem-solving, while refining my communication and leadership abilities. This role was instrumental in developing a high level of time management and the capacity to coordinate team efforts to ensure the success of complex projects.
- Since my first year of university (A.Y. 2020–2021), I have provided academic tutoring to high school and university students, assisting them with exercises and conceptual explanations to deepen their understanding of STEM subjects. My approach focuses on training essential skills such as problem-solving, critical thinking, and independent study management, helping students develop autonomous and effective learning strategies.

Projects

Master thesis project (still in progress) *University of Milan*

Black Holes with Homogeneous Horizon in Gauged Supergravity *2025–2026*

- Finding solutions for 5D black holes with homogeneous horizon, with possible extensions to the 8 Thurston geometries. The work is mainly developed in 5D AdS and asymptotically AdS settings.

Astrophysics Data Analysis *University of Milano - Bicocca*

Numerical Simulations in Python *2025*

- Developed numerical models to study astrophysical systems and visualize results with scientific plots, including machine-learning-based techniques.
- Used Python tools for data analysis, validation, and interpretation of simulation outcomes.

Harmonic Oscillator and Wilson Cluster *University of Milano - Bicocca*

Numerical Simulations in C *2025–2026*

- Explored advanced C simulations of a quantum harmonic oscillator, with the goal of computing the mass and ground-state energy.
- Used the Wilson Cluster (Milano - Bicocca) to generate data for numerical computation of topological susceptibility in SU(3) Yang–Mills theory and the chiral anomaly.

Research Interests

My research interests lie primarily in theoretical physics, driven by both foundational questions and the mathematical challenges underlying them. During my master's thesis, I solved Einstein equations for various black hole solutions, mainly focusing on static, rotating, and charged cases in AdS_5. To support this research, I built a solid foundation in String Theory and Supergravity (SUGRA). Moving forward, I am eager to pursue further research in these areas. My main interests include:

- Quantum field theory
- AdS/CFT correspondence and string theory
- General relativity
- Numerical methods in theoretical physics

Languages

Italian	Native
English	Professional working proficiency (B2/C1)

Extracurricular Activity

In my free time, I practice rock climbing, running, and strength training. I have a deep interest in literature (both scientific and non-scientific), videogames, and traveling. I usually travel 2–3 times a year and have visited many places in Europe, as well as some non-European countries.

I hereby authorize the processing of my personal data in accordance with EU Regulation 2016/679 (GDPR) for recruitment and personnel selection purposes.